I) LP-Rounding



Consider a minimization problem Π in ILP-form.

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Approximation factor $ALG/OPT_{\Pi} \leq ALG/OPT_{relax}$

II) Primal-Dual Approach



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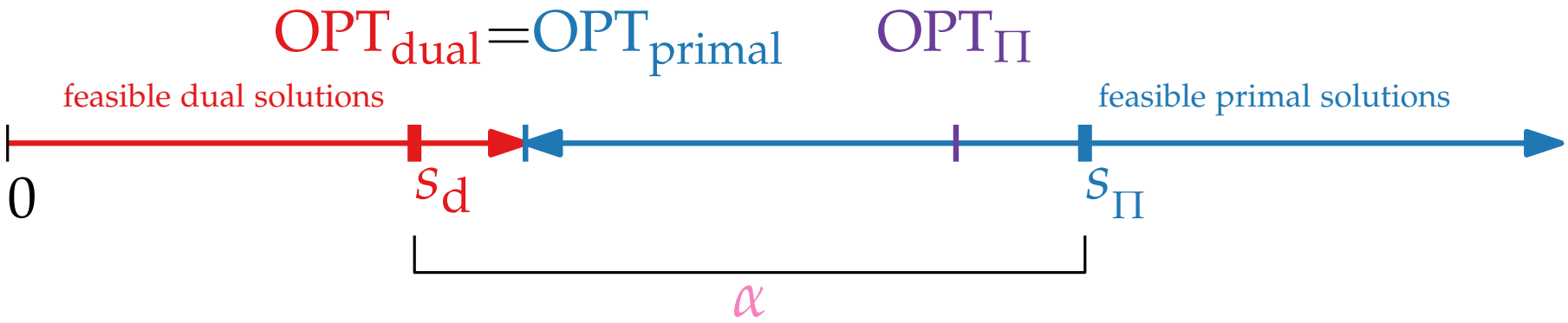


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Approximation factor $\leq \text{obj}(s_{\Pi}) / \text{obj}(s_d)$

III) Dual Fitting



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Combinatorial algorithm (e.g., greedy) computes feasible primal solution s_{Π} and infeasible dual solution s_d which "completely pays for" s_{Π},

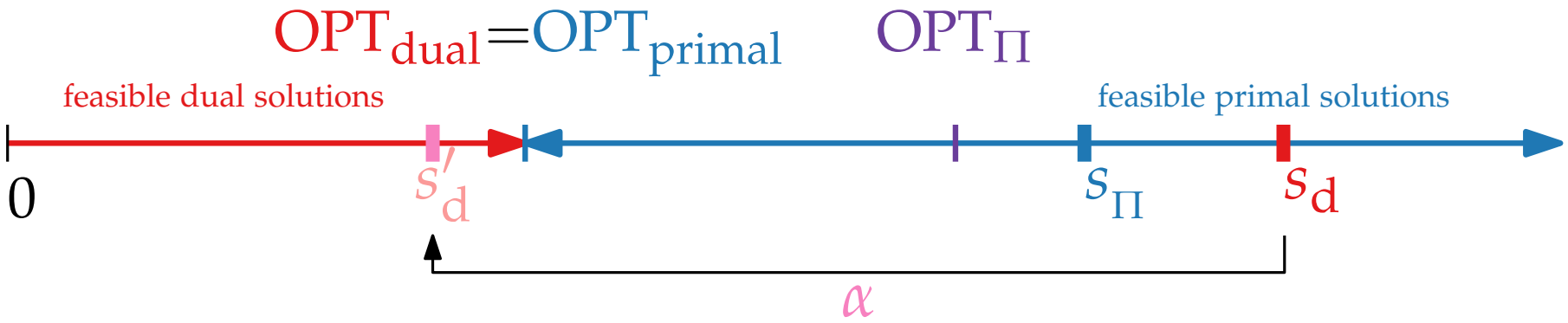
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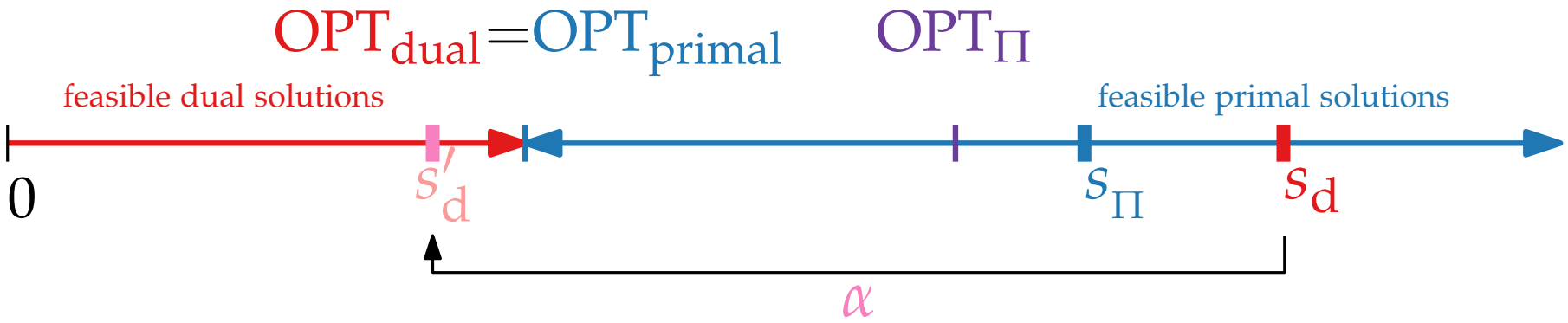


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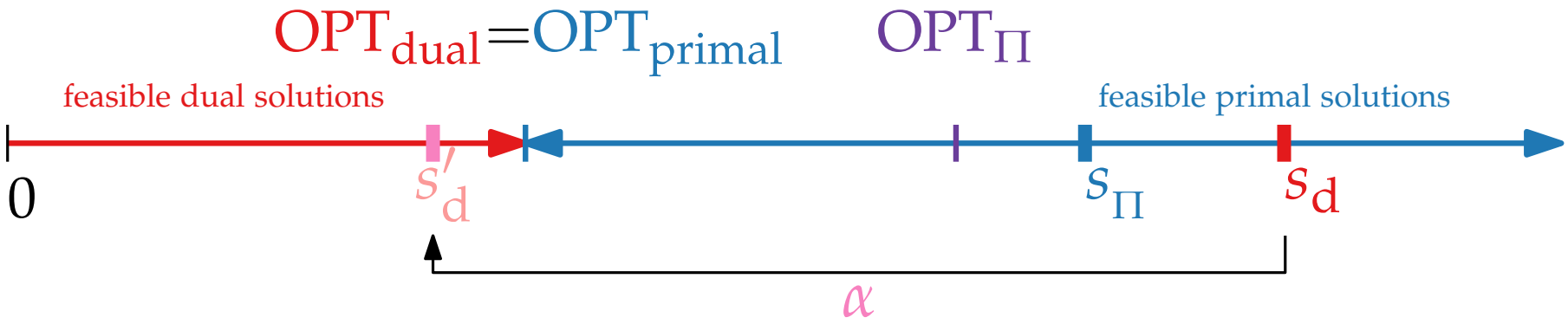
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$\Rightarrow$ $\mathrm{obj}(s'_{\mathrm{d}}) \leq \mathrm{OPT}_{\mathrm{dual}} \leq \mathrm{OPT}_{\Pi}$

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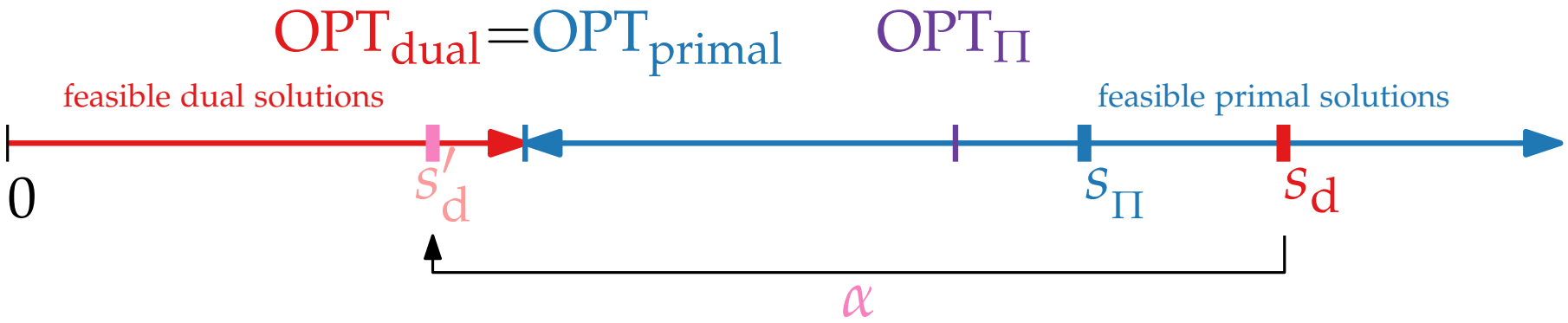
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$$\Rightarrow \qquad \text{obj}(s_d)/\alpha = \text{obj}(s'_d) \leq OPT_{dual} \leq OPT_{\Pi}$$

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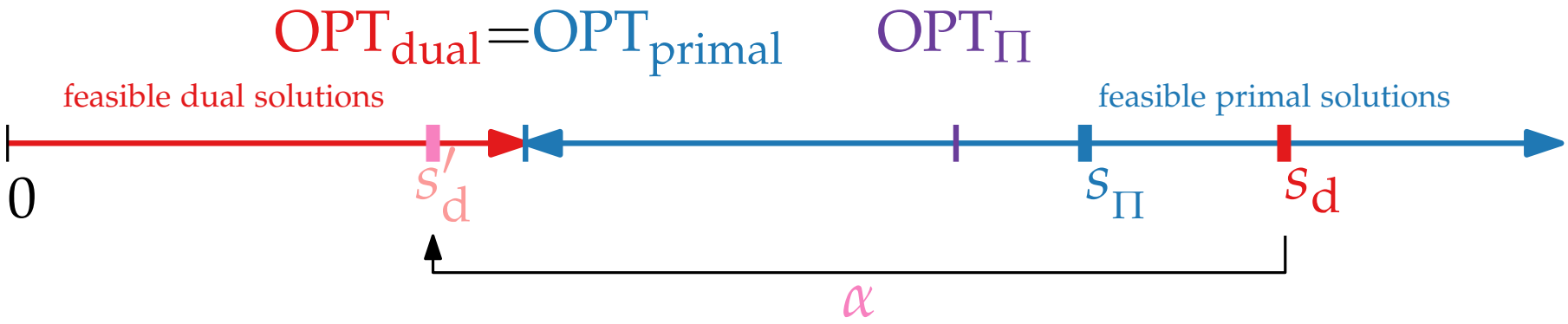
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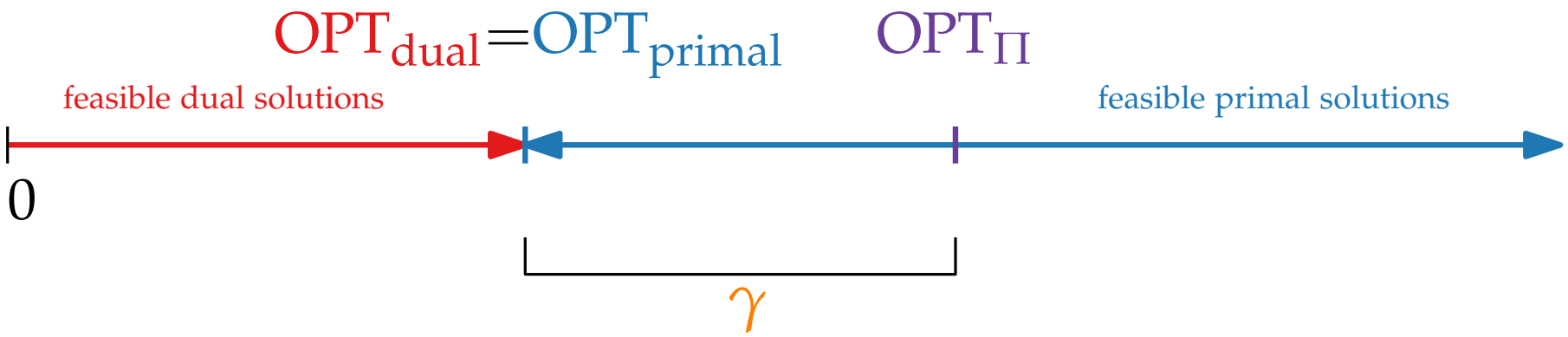
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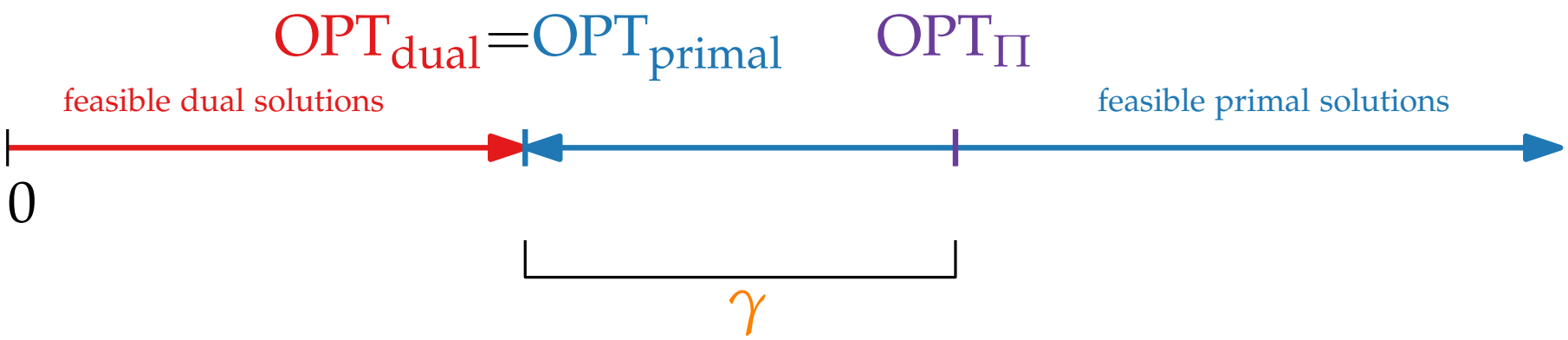
$\Rightarrow$ Scaling factor α is approximation factor.

Integrality Gap



Consider a minimization problem Π in ILP-form.

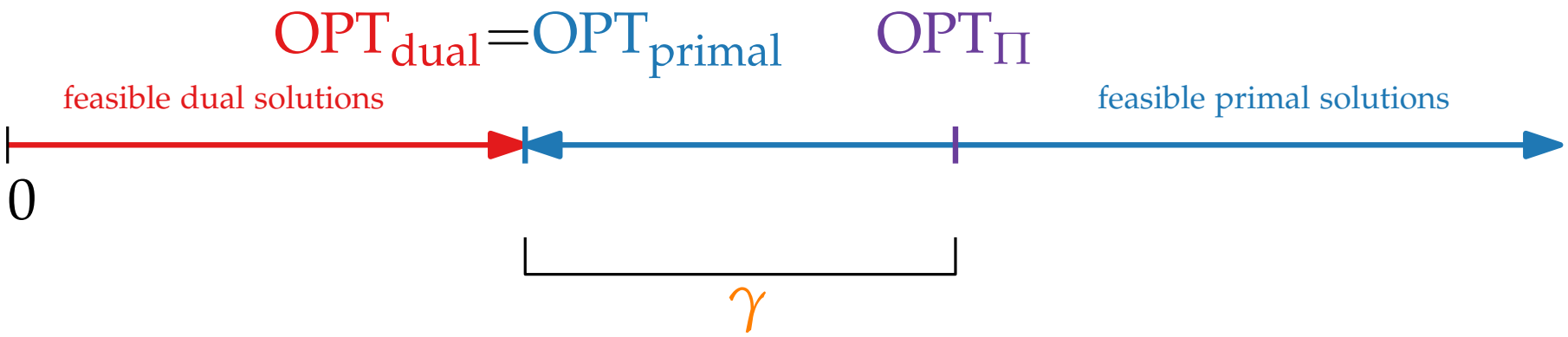
Integrality Gap



Consider a minimization problem Π in ILP-form.

Dual methods (without outside help) are limited by the *Integrality Gap* of the LP-relaxation

Integrality Gap

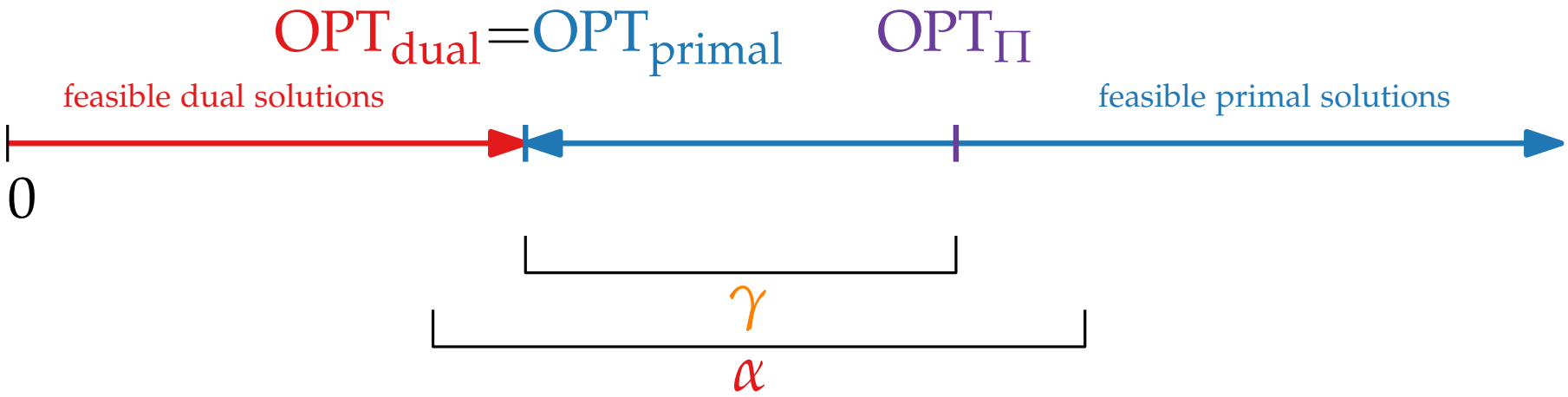


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$$\therefore \gamma = \sup_{I} \frac{\mathrm{OPT}_{\Pi}(I)}{\mathrm{OPT}_{\mathrm{primal}}(I)}$$

Integrality Gap



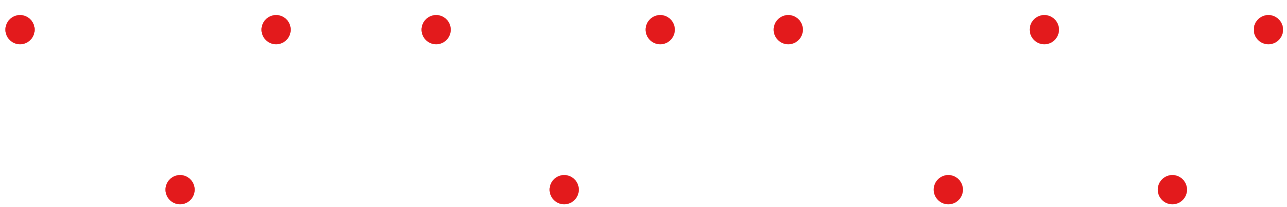
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$$\alpha \geq \gamma = \sup_{I} \frac{\mathrm{OPT}_{\Pi}(I)}{\mathrm{OPT}_{\mathrm{primal}}(I)}$$

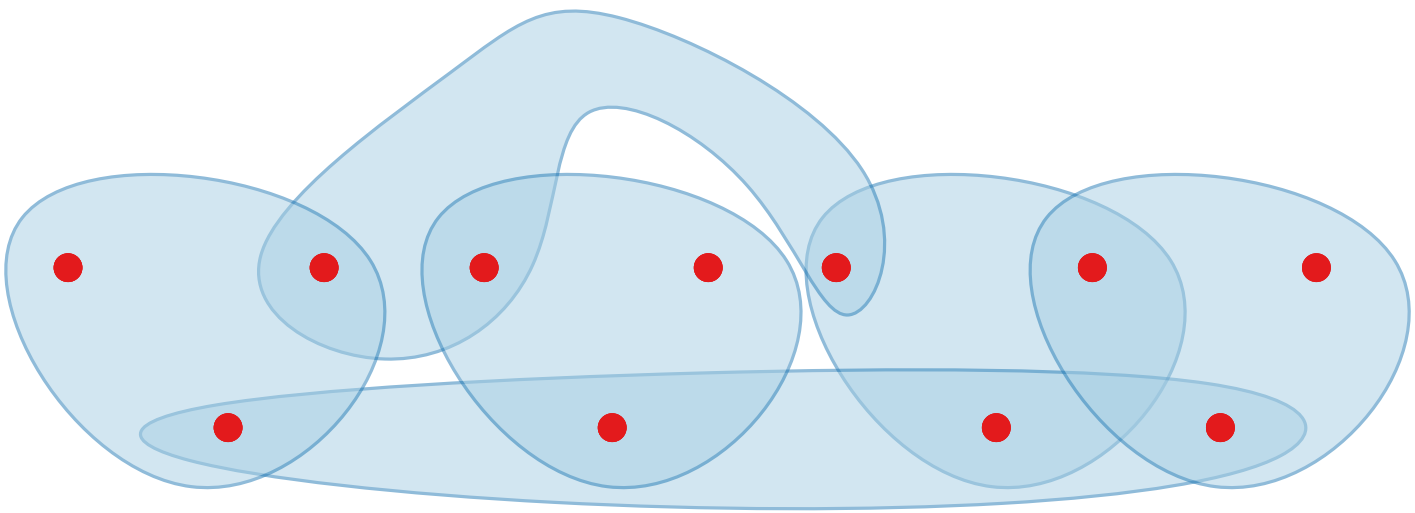
SetCover as an ILP

Ground set U



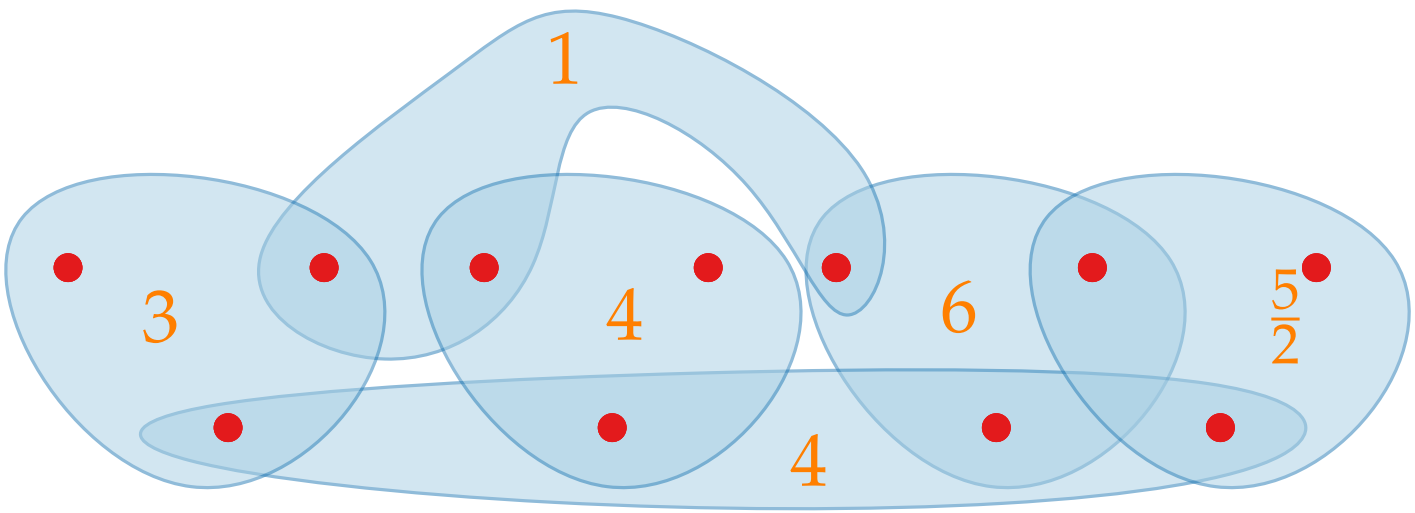
SetCover as an ILP

Ground set U
Family $\mathcal{S} \subseteq 2^U$ with $\bigcup \mathcal{S} = U$



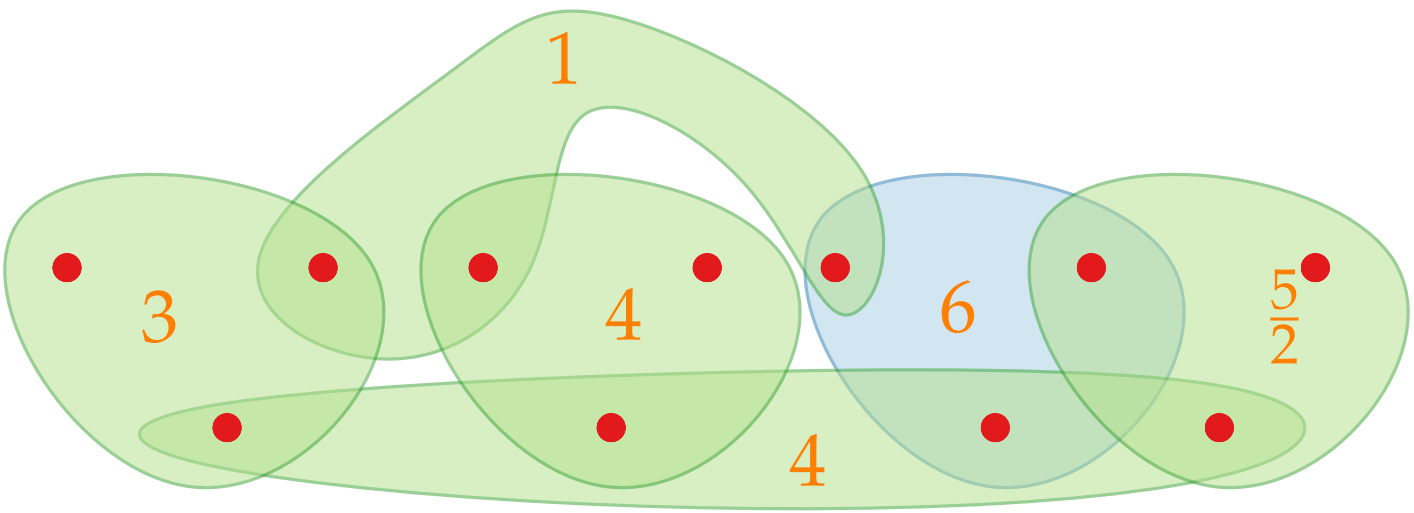
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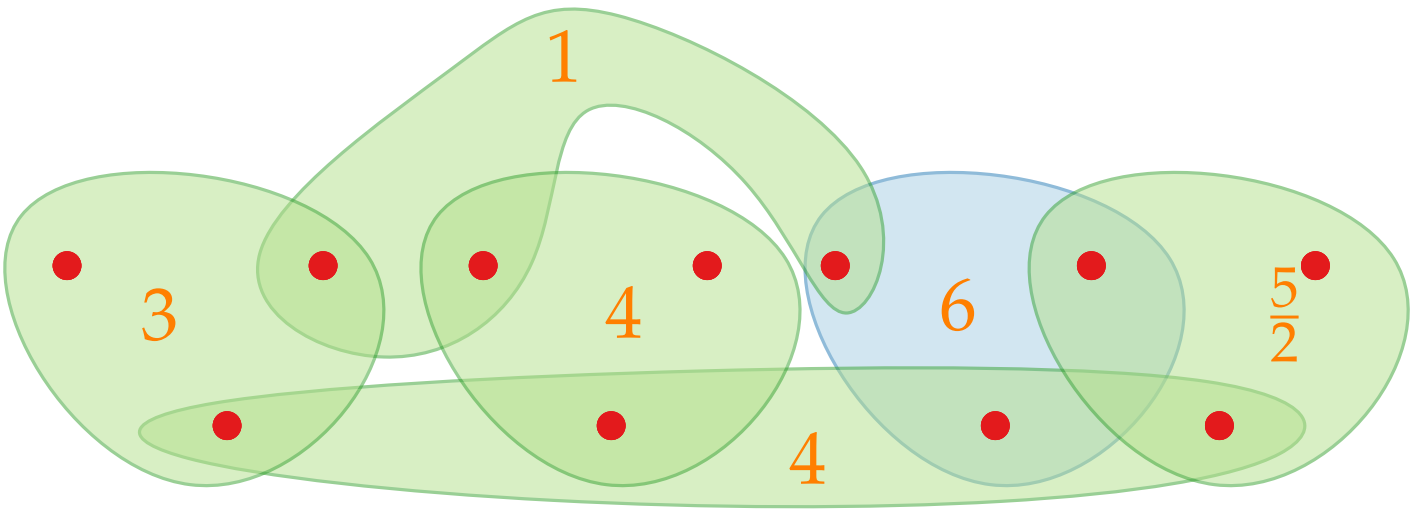
Find cover $\mathcal{S}' \subseteq \mathcal{S}$ of U with minimum cost.

SetCover as an ILP

minimize

subject to

Ground set U
Family $\mathcal{S} \subseteq 2^U$ with $\bigcup \mathcal{S} = U$
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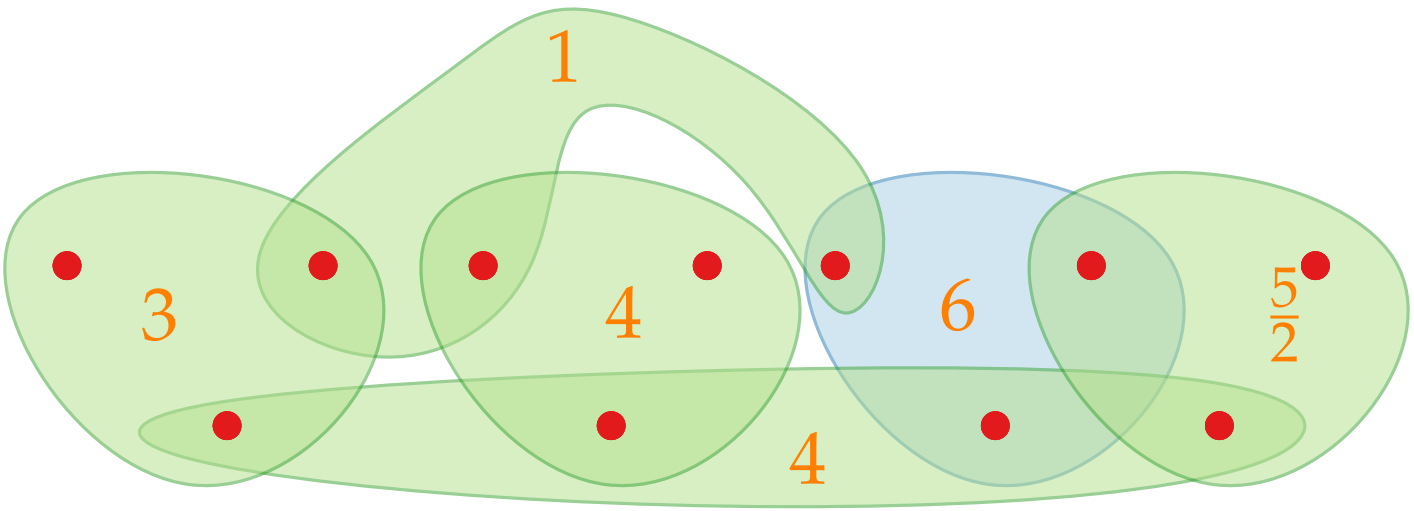
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x_S $S \in \mathcal{S}$

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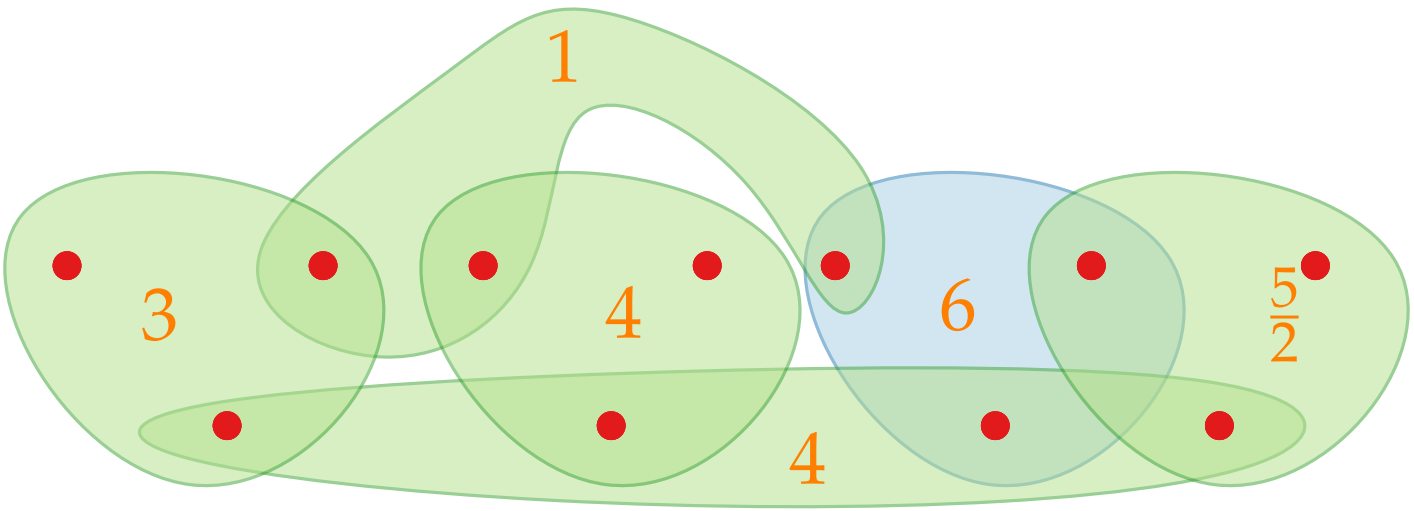
SetCover as an ILP

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$$x_S \in \{0,1\} \quad S \in \mathcal{S}$$

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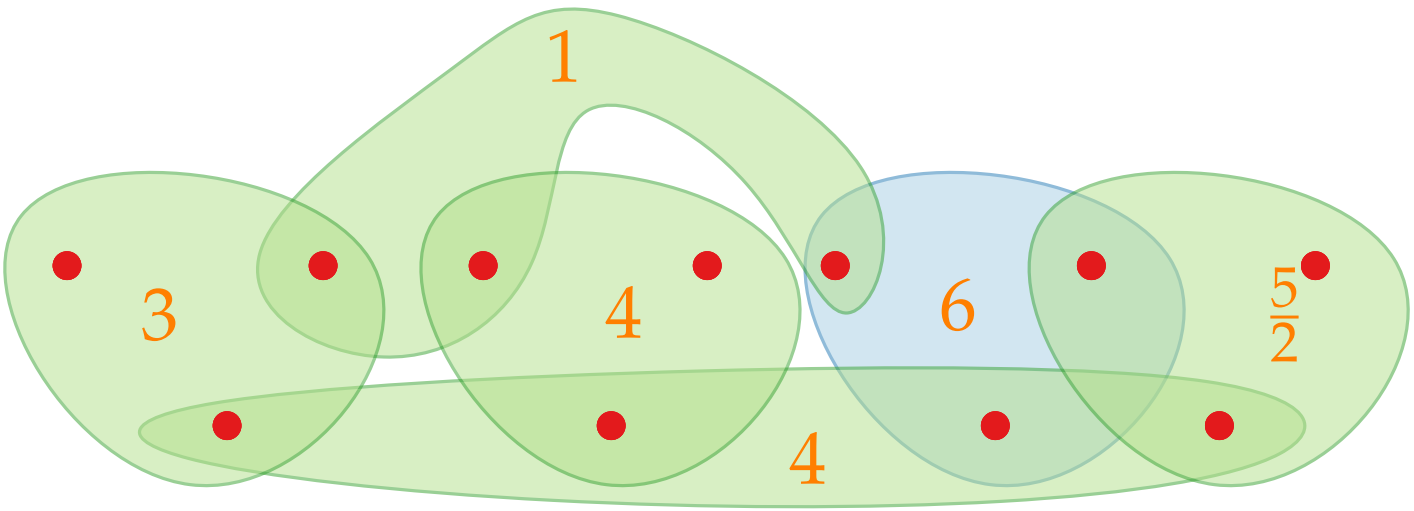
SetCover as an ILP

minimize $\sum_{S \in \mathcal{S}} c_S x_S$

subject to

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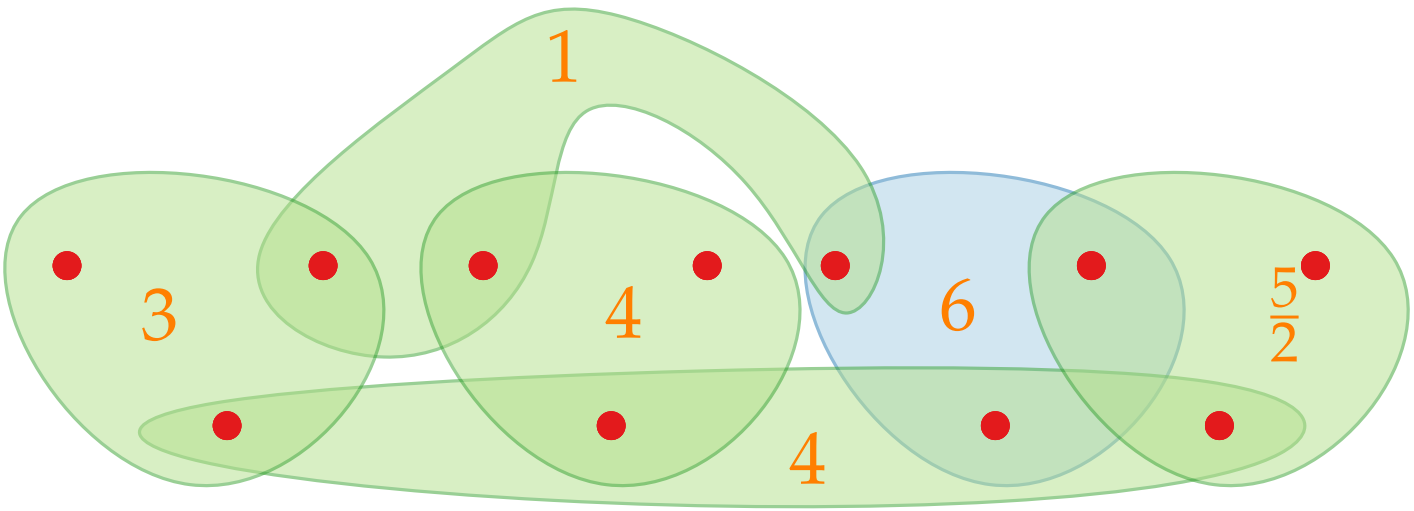


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SetCover as an ILP

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \in \{0,1\} \qquad S \in \mathcal{S} \end{aligned}$$

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SetCover – LP-Relaxation

$$\begin{aligned}
&\textbf{minimize} && \sum_{S \in \mathcal{S}} c_S x_S && \\
&\textbf{subject to} && \sum_{S \ni u} x_S \geq 1 && u \in U \\
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\end{aligned}$$

SetCover – LP-Relaxation

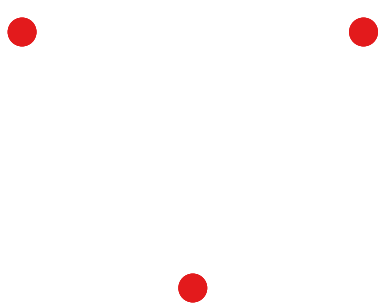
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Optimal?

SetCover – LP-Relaxation

$$\begin{aligned} &\textbf{minimize} && \sum_{S \in \mathcal{S}} c_S x_S && \\ &\textbf{subject to} && \sum_{S \ni u} x_S \geq 1 && u \in U \\ & && x_S \geq 0 && S \in \mathcal{S} \end{aligned}$$

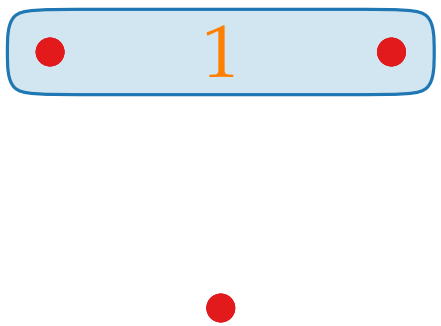
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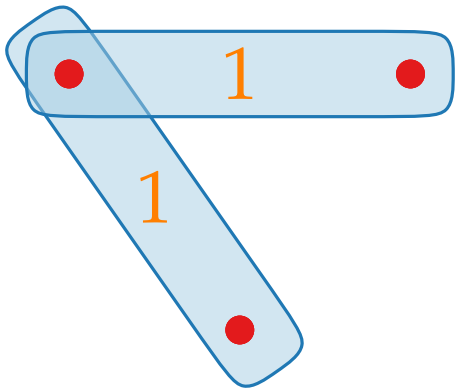
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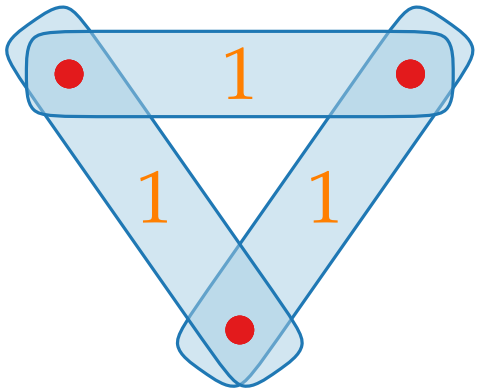
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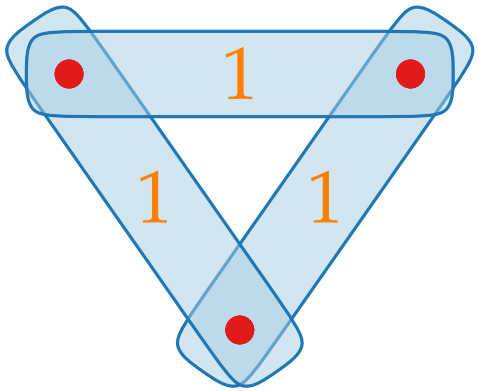
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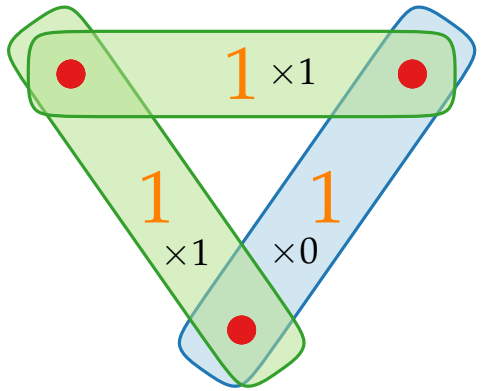


SetCover – LP-Relaxation

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Optimal?



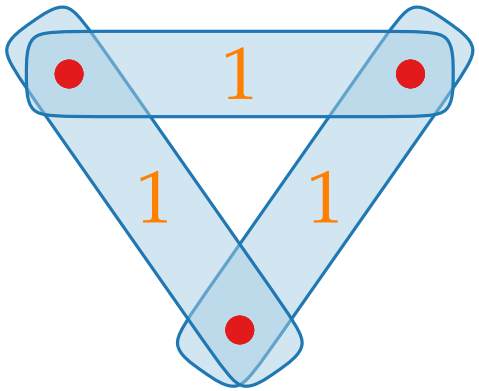


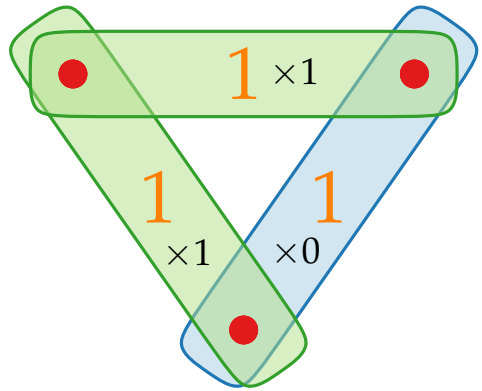
integer: 2

SetCover – LP-Relaxation

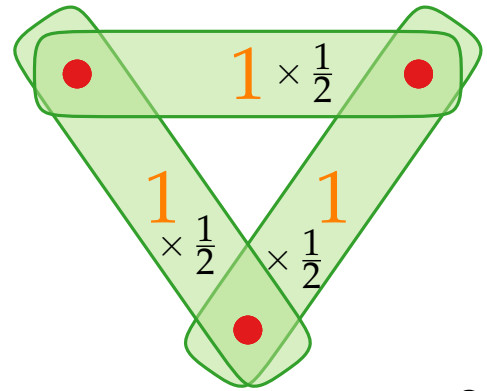
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Optimal?





integer: 2



fractional: $\frac{3}{2}$

SetCover – Dual LP

minimize $\sum_{S \in \mathcal{S}} c_S x_S$

subject to $\sum_{S \ni u} x_S \geq 1 \qquad u \in U$

$x_S \geq 0 \qquad S \in \mathcal{S}$

SetCover – Dual LP

minimize $\sum_{S \in \mathcal{S}} c_S x_S$

subject to $\sum_{S \ni u} x_S \geq 1 \qquad u \in U$

$x_S \geq 0 \qquad S \in \mathcal{S}$

maximize

subject to

SetCover – Dual LP

minimize $\sum_{S \in \mathcal{S}} c_S x_S$

subject to $\sum_{S \ni u} x_S \geq 1 \qquad u \in U$

$x_S \geq 0 \qquad S \in \mathcal{S}$

maximize

subject to

$y_u \geq 0 \qquad u \in U$

SetCover – Dual LP

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

$$\begin{aligned} \textbf{maximize} \quad & \sum_{u \in U} y_u \\ \textbf{subject to} \quad & \\ & y_u \geq 0 \qquad u \in U \end{aligned}$$

SetCover – Dual LP

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

$$\begin{aligned} \textbf{maximize} \quad & \sum_{u \in U} y_u \\ \textbf{subject to} \quad & \sum_{u \in S} y_u \leq c_S \qquad S \in \mathcal{S} \\ & y_u \geq 0 \qquad u \in U \end{aligned}$$

LP-Rounding: Approach I

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

LP-Rounding-One($U, \mathcal{S}, c$)

LP-Rounding: Approach I

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

LP-Rounding-One($U, \mathcal{S}, c$)

Compute optimal solution x for LP-Relaxation.
Round each x_S with $x_S > 0$ to 1.

LP-Rounding: Approach I

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

LP-Rounding-One($U, \mathcal{S}, c$)

Compute optimal solution x for LP-Relaxation.
Round each x_S with $x_S > 0$ to 1.

Generates a valid solution

LP-Rounding: Approach I

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

LP-Rounding-One($U, \mathcal{S}, c$)

Compute optimal solution x for LP-Relaxation.
Round each x_S with $x_S > 0$ to 1.

Generates a valid solution

Scaling factor arbitrarily large

LP-Rounding: Approach I

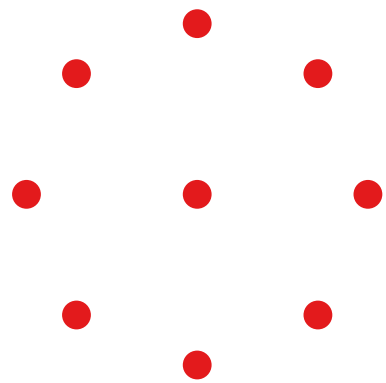
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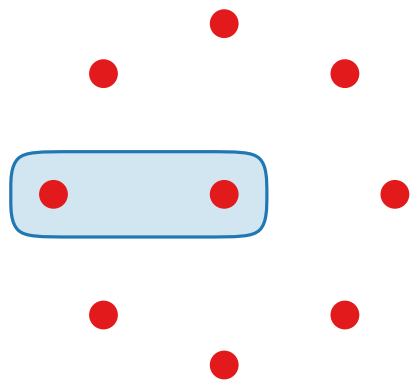
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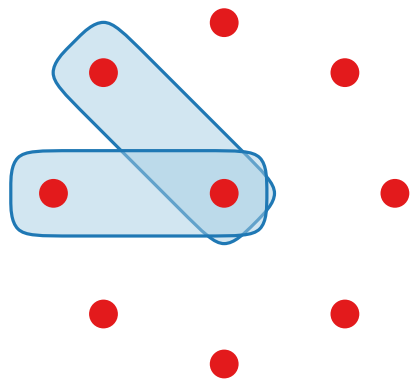
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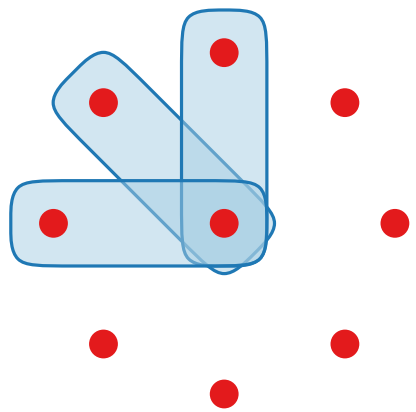
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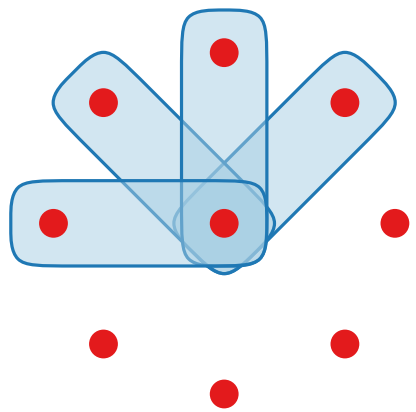
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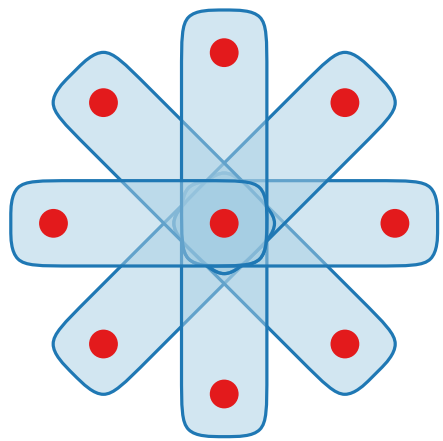
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Generates a valid solution

Scaling factor arbitrarily large



LP-Rounding: Approach I

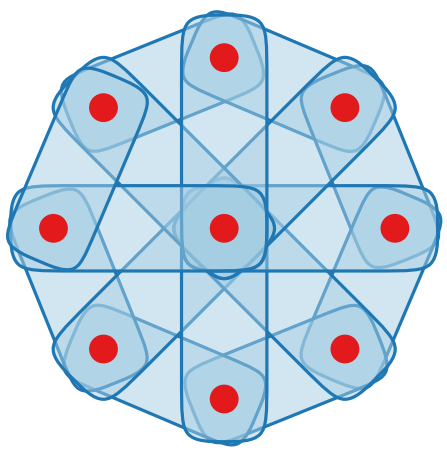
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LP-Rounding-One($U, \mathcal{S}, c$)

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Generates a valid solution

Scaling factor arbitrarily large



LP-Rounding: Approach I

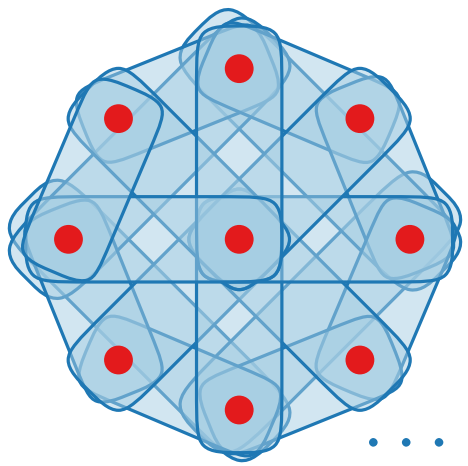
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LP-Rounding: Approach I

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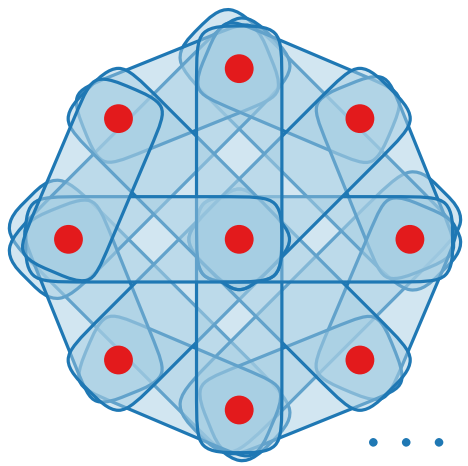
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Compute optimal solution x for LP-Relaxation.
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Generates a valid solution

Scaling factor arbitrarily large

Use frequency f



LP-Rounding: Approach II

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

LP-Rounding-Two($U, \mathcal{S}, c$)

Compute optimal solution x for LP-Relaxation.
Round each x_S with $x_S \geq 1/f$ to 1; remaining to 0.

Let f be the frequency of (number of sets containing) the most frequent element.

LP-Rounding: Approach II

$$\begin{aligned} \textbf{minimize} \quad & \sum_{S \in \mathcal{S}} c_S x_S \\ \textbf{subject to} \quad & \sum_{S \ni u} x_S \geq 1 \qquad u \in U \\ & x_S \geq 0 \qquad S \in \mathcal{S} \end{aligned}$$

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Theorem. LP-Rounding-Two is a factor-f-approximation algorithm for SetCover.

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Relaxing Complementary Slackness

minimize	$c^\top x$		
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	x	$\geq$	0

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~~**Primal CS**~~: Relaxed Primal CS

For each $j = 1, \ldots, n$: either $x_j = 0$ or ~~$\sum_{i=1}^{m} a_{ij} y_i = c_j$~~

$$c_j / \alpha \leq \sum_{i=1}^{m} a_{ij} y_i \leq c_j$$

Dual CS:

For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

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minimize	$c^\top x$		
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~~**Dual CS**~~: Relaxed Dual CS

For each $i = 1, \ldots, m$: either $y_i = 0$ or ~~$\sum_{j=1}^n a_{ij} x_j = b_i$~~

$$b_i \leq \sum_{j=1}^n a_{ij} x_j \leq \beta \cdot b_i$$

$$\Leftrightarrow \sum_{j=1}^n c_j x_j = \sum_{i=1}^m b_i y_i$$

Relaxing Complementary Slackness

minimize	$c^\intercal x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\intercal y$		
subject to	$A^\intercal y$	$\leq$	c
	y	$\geq$	0

~~**Primal CS**~~: Relaxed Primal CS

For each $j = 1, \ldots, n$: either $x_j = 0$ or ~~$\sum_{i=1}^{m} a_{ij} y_i = c_j$~~

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~~**Dual CS**~~: Relaxed Dual CS

For each $i = 1, \ldots, m$: either $y_i = 0$ or ~~$\sum_{j=1}^{n} a_{ij} x_j = b_i$~~

$$b_i \leq \sum_{j=1}^{n} a_{ij} x_j \leq \beta \cdot b_i$$

$$\Leftrightarrow \sum_{j=1}^{n} c_j x_j = \sum_{i=1}^{m} b_i y_i \quad \Rightarrow \sum_{j=1}^{n} c_j x_j \leq \alpha\beta \sum_{i=1}^{m} b_i y_i \leq \alpha\beta \cdot \mathrm{OPT}_{\mathrm{LP}}$$

Primal-Dual Schema

Start with a feasible dual and infeasible primal solution (often trivial).

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Start with a feasible dual and infeasible primal solution (often trivial).

"Improve" the feasibility of the primal solution...

...and simultaneously the obj. value of the dual solution.

Do so until the relaxed CS conditions are met.

Maintain that the primal solution is integer valued.

The feasibility of the primal solution and relaxed CS condition provide an approximation ratio.

Relaxed CS for SetCover

minimize $\sum_{S \in \mathcal{S}} c_S x_S$

subject to $\sum_{S \ni u} x_S \geq 1 \qquad u \in U$

$x_S \geq 0 \qquad S \in \mathcal{S}$

maximize $\sum_{u \in U} y_u$

subject to $\sum_{u \in S} y_u \leq c_S \qquad S \in \mathcal{S}$

$y_u \geq 0 \qquad u \in U$

Relaxed CS for SetCover

minimize	$\sum_{S \in \mathcal{S}} c_S x_S$	
subject to	$\sum_{S \ni u} x_S \geq 1$	$u \in U$
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maximize	$\sum_{u \in U} y_u$	
subject to	$\sum_{u \in S} y_u \leq c_S$	$S \in \mathcal{S}$
	$y_u \geq 0$	$u \in U$

(Unrelaxed) primal CS:

Relaxed CS for SetCover

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subject to	$\sum_{S \ni u} x_S \geq 1$	$u \in U$	
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maximize	$\sum_{u \in U} y_u$		
subject to	$\sum_{u \in S} y_u \leq c_S$	$S \in \mathcal{S}$	
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(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow$

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subject to	$\sum_{u \in S} y_u \leq c_S$	$S \in \mathcal{S}$	
	$y_u \geq 0$	$u \in U$	

(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow \sum_{u \in S} y_u = c_S$

Relaxed CS for SetCover

minimize	$\sum_{S \in \mathcal{S}} c_S x_S$		**maximize**	$\sum_{u \in U} y_u$	
subject to	$\sum_{S \ni u} x_S \geq 1$	$u \in U$	**subject to**	$\sum_{u \in S} y_u \leq c_S$	$S \in \mathcal{S}$
	$x_S \geq 0$	$S \in \mathcal{S}$		$y_u \geq 0$	$u \in U$

critical set

(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow \sum_{u \in S} y_u = c_S$

Relaxed CS for SetCover

minimize	$\sum_{S \in \mathcal{S}} c_S x_S$		
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(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow \sum_{u \in S} y_u = c_S$

critical set

only chooses critical sets

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critical set

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Relaxed dual CS:

Relaxed CS for SetCover

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maximize	$\sum_{u \in U} y_u$			
subject to	$\sum_{u \in S} y_u \leq c_S$	$S \in \mathcal{S}$		
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critical set

(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow \sum_{u \in S} y_u = c_S$

only chooses critical sets

Relaxed dual CS: $y_u \neq 0 \Rightarrow$

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(Unrelaxed) primal CS: $x_S \neq 0 \Rightarrow \sum_{u \in S} y_u = c_S$

critical set

only chooses critical sets

Relaxed dual CS: $y_u \neq 0 \Rightarrow 1 \leq \sum_{S \ni u} x_S \leq f$.

Relaxed CS for SetCover

minimize	$\sum_{S \in \mathcal{S}} c_S x_S$		
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	$x_S \geq 0$	$S \in \mathcal{S}$	

maximize	$\sum_{u \in U} y_u$		
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only chooses critical sets

trivial for binary x

Relaxed dual CS: $y_u \neq 0 \Rightarrow 1 \leq \sum_{S \ni u} x_S \leq f \cdot 1$

Primal-Dual-Schema for SetCover

PrimalDualSetCover($U, \mathcal{S}, c$)

$x \leftarrow 0, y \leftarrow 0$

repeat

|

|

|

until all elements are covered.

return x

Primal-Dual-Schema for SetCover

PrimalDualSetCover($U, \mathcal{S}, c$)

$x \leftarrow 0, y \leftarrow 0$

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  Select an uncovered element u.

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  Increase y_u until a set S is critical ($\sum_{u' \in S} y_{u'} = c_S$).

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  Mark all elements in these sets as covered.

until all elements are covered.

return x

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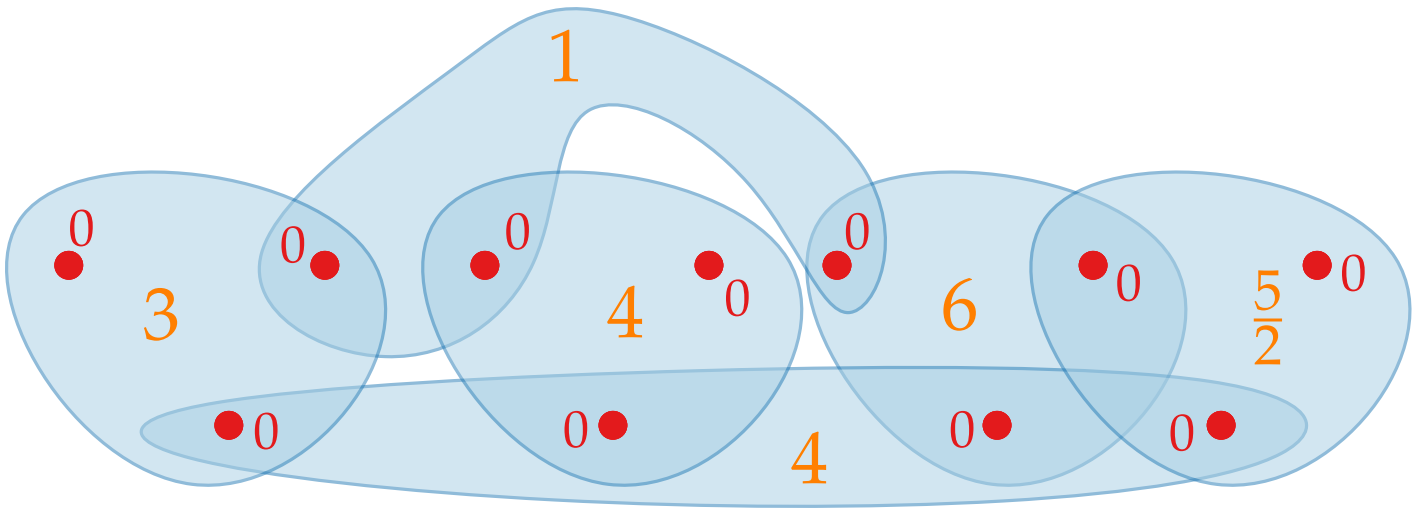
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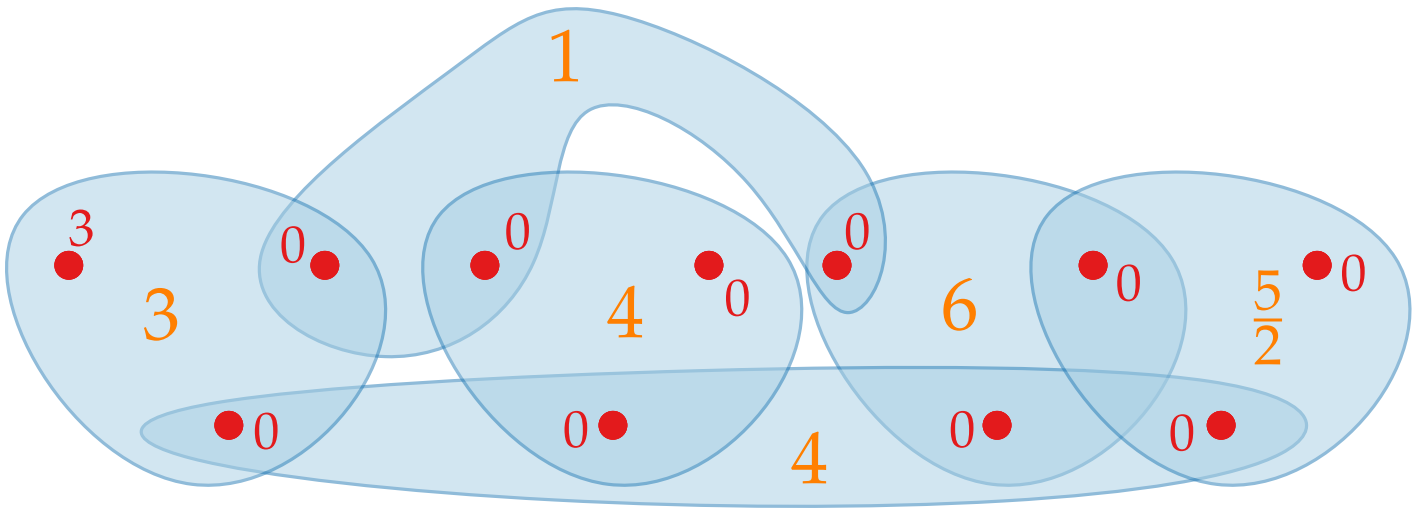
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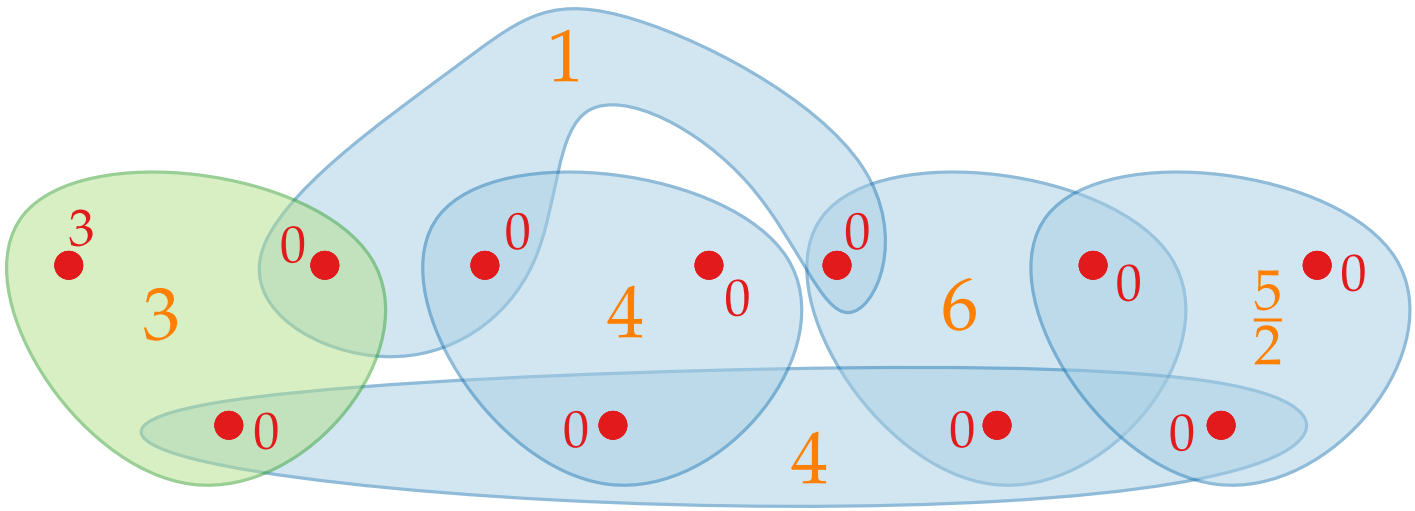
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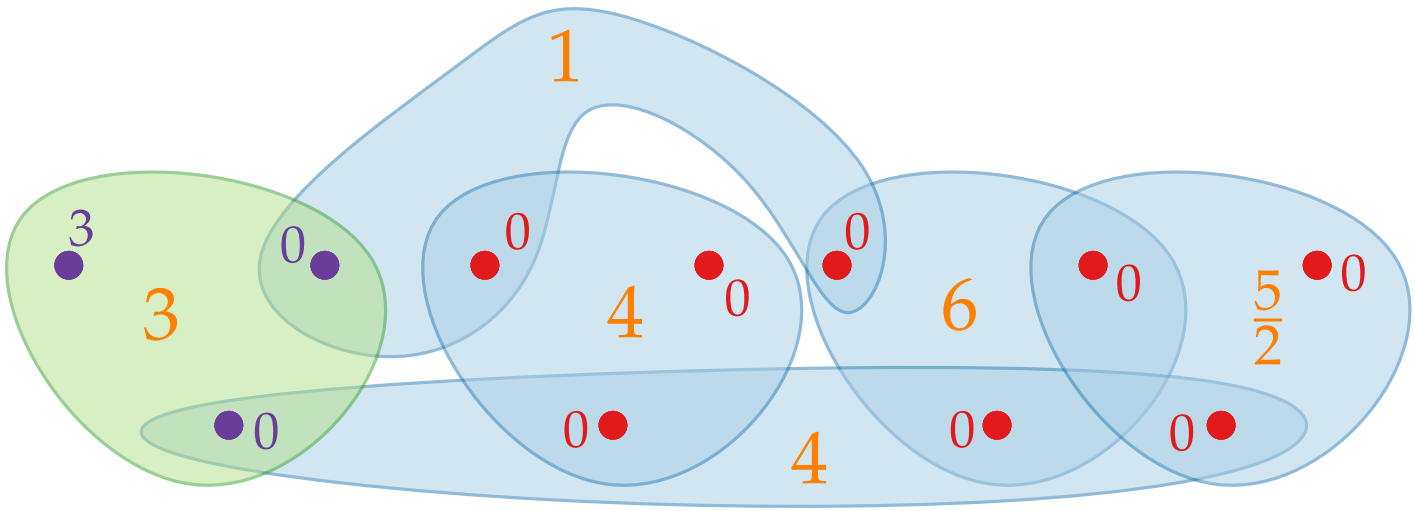
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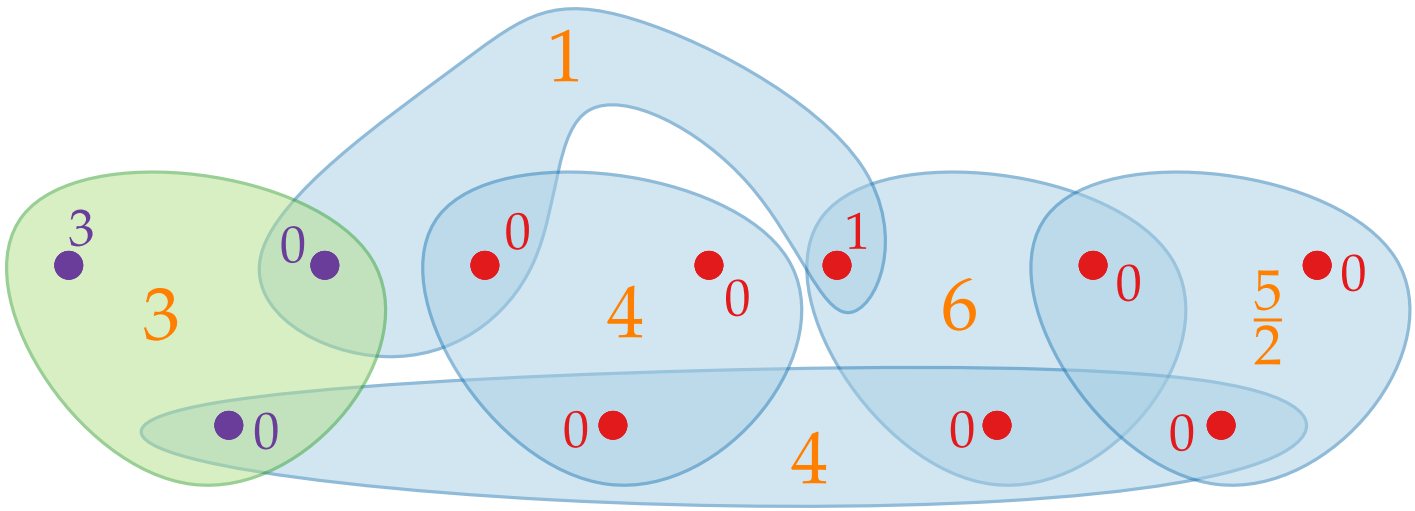
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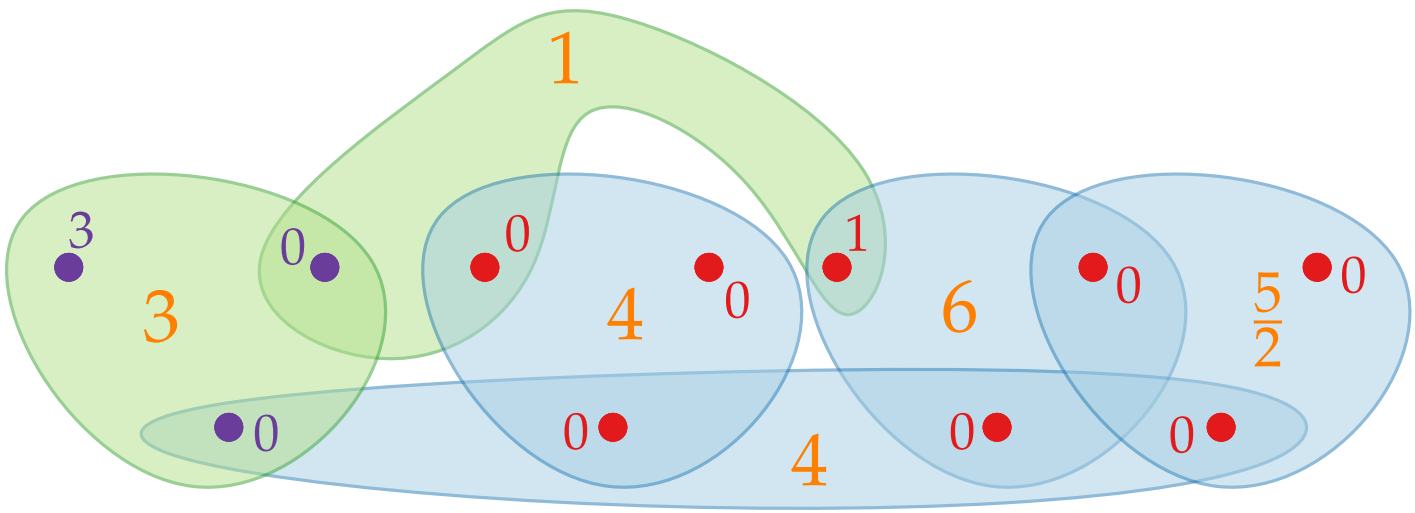
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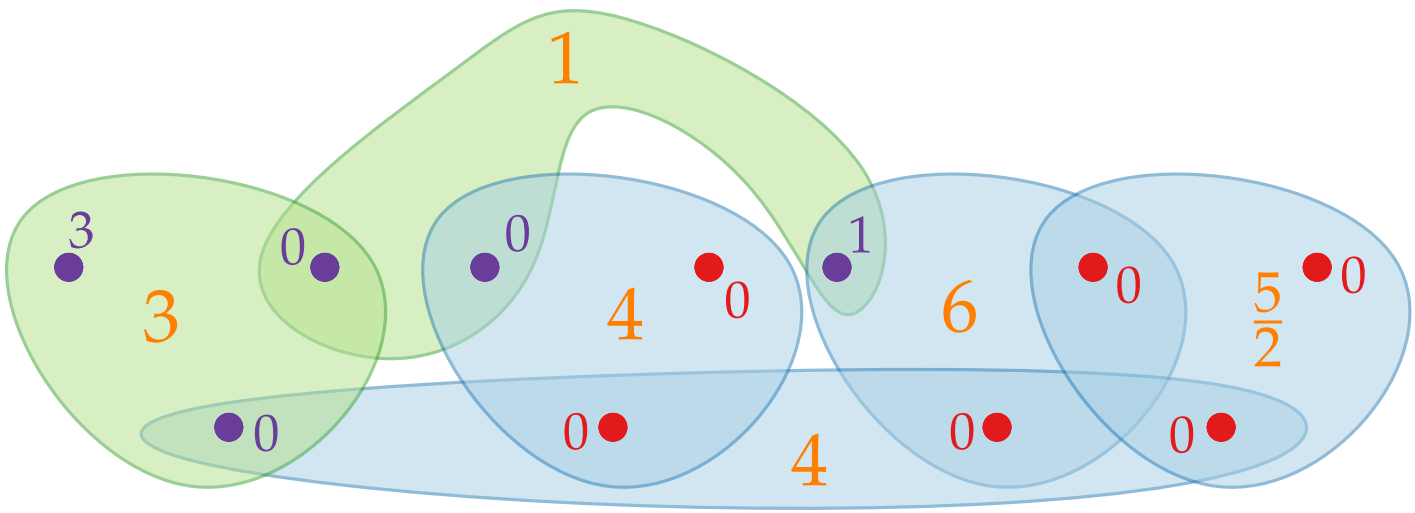
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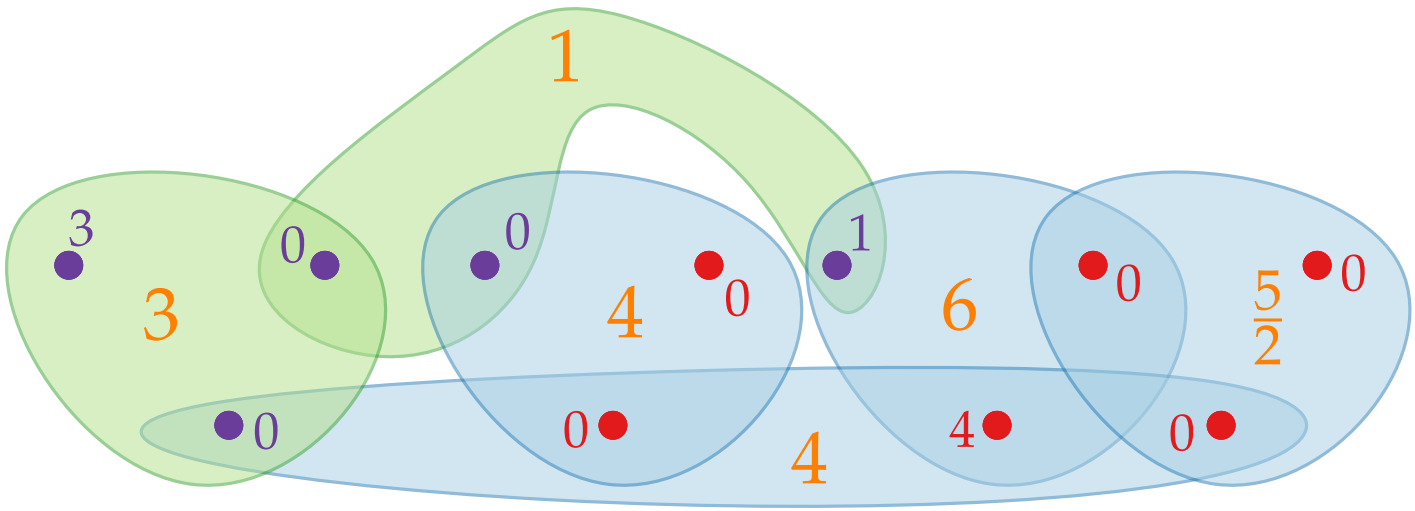
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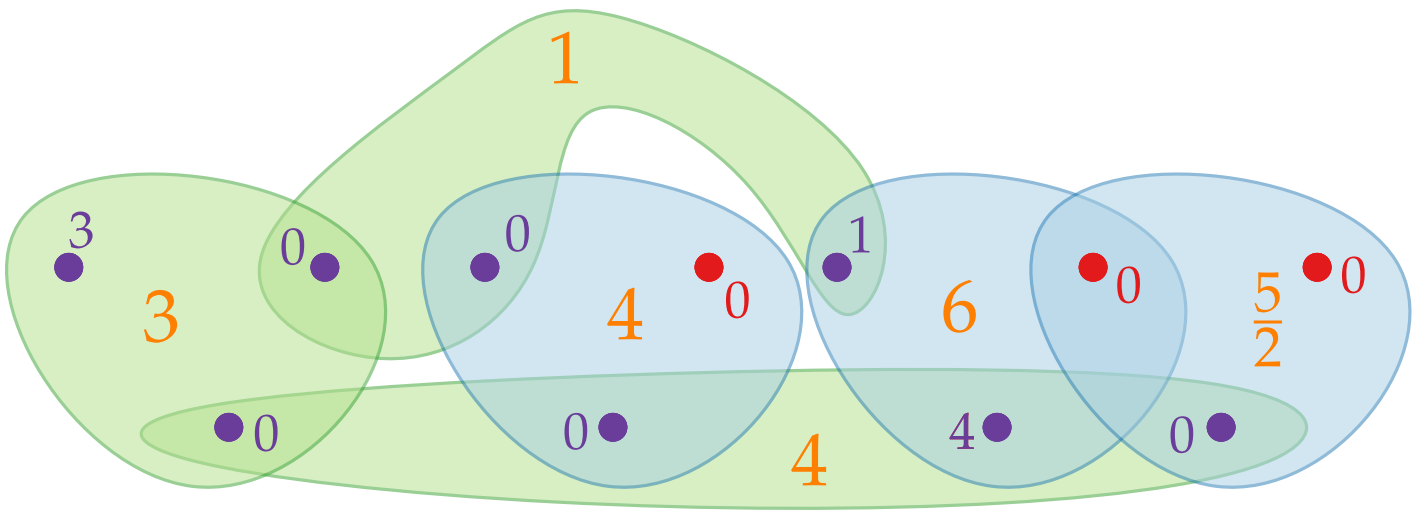
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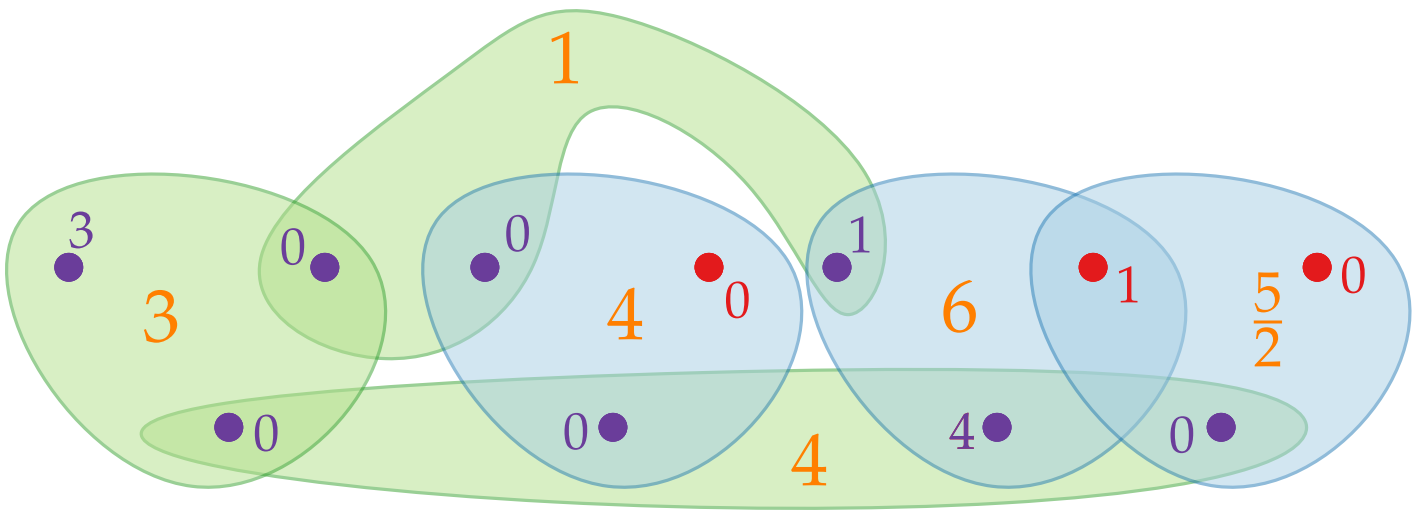
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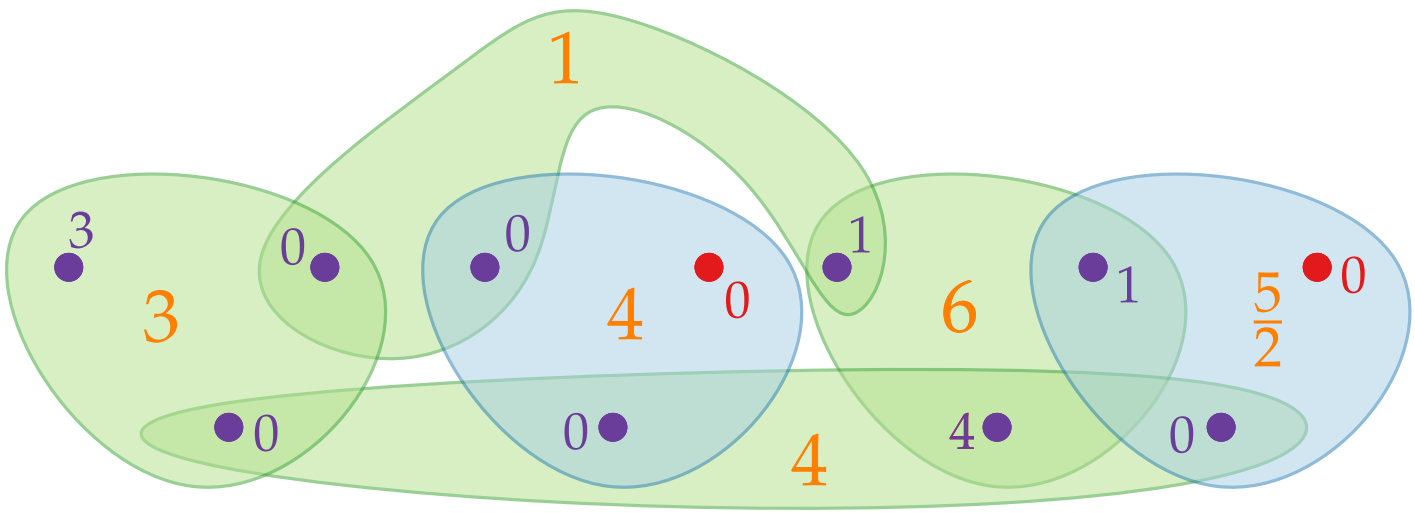
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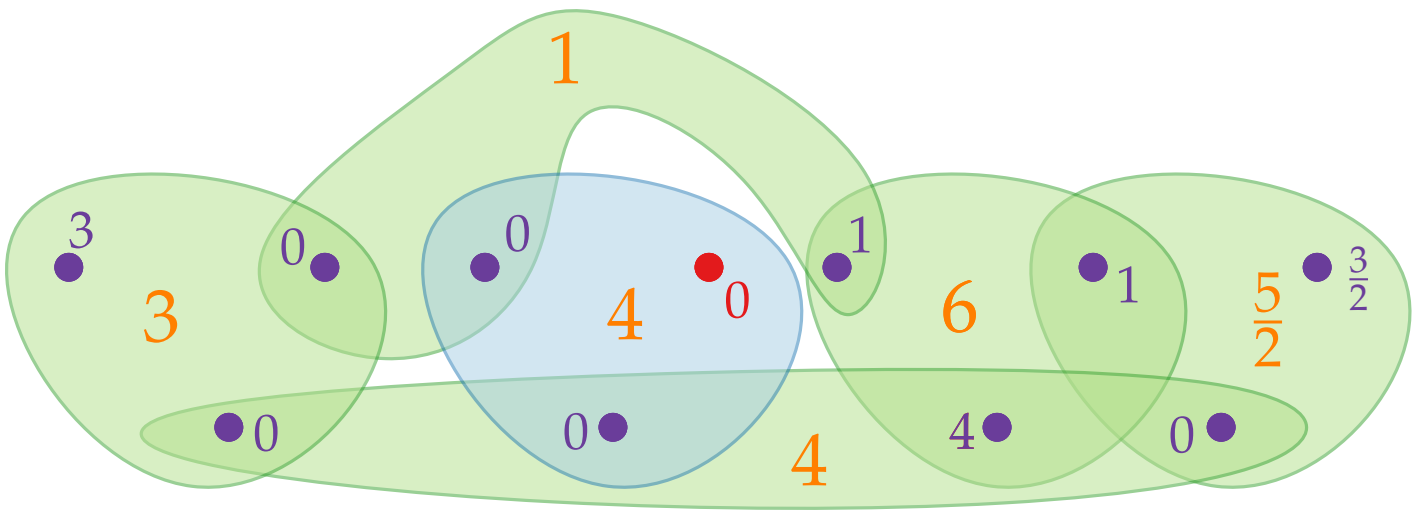
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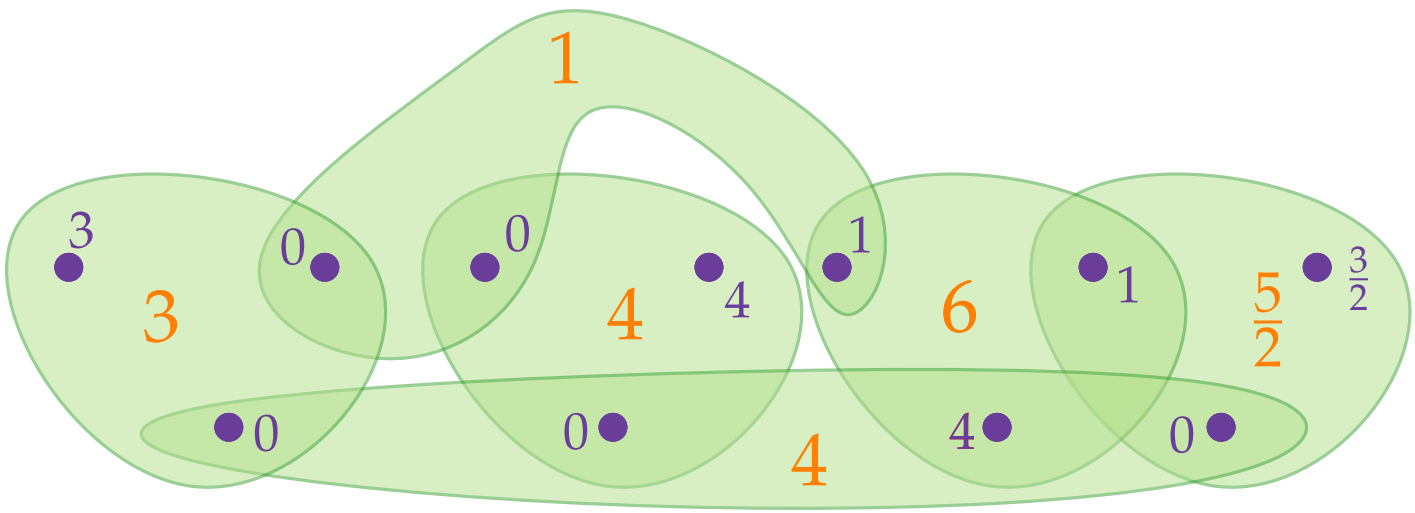
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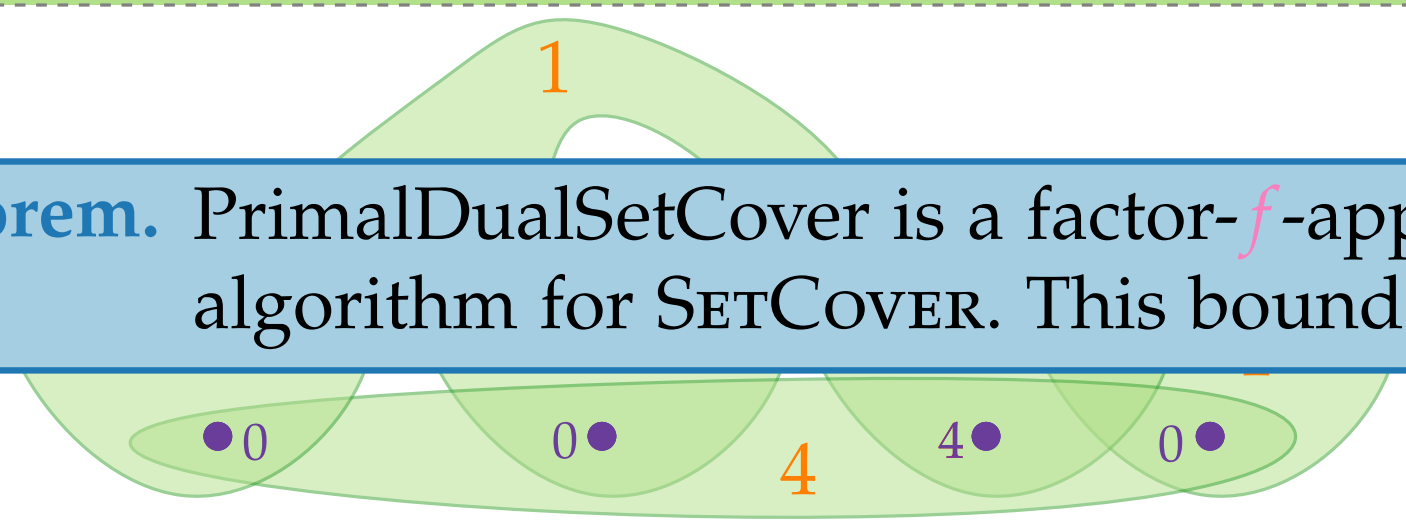
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Theorem. PrimalDualSetCover is a factor-f-approximation algorithm for SetCover. This bound is tight.



Dual Fitting for SetCover

Combinatorial (greedy) algorithm (see Lecture 02):

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GreedySetCover(U, S, c)
    C ← ∅
    S' ← ∅
    while C ≠ U do
        S ← Set from S that minimizes c(S) / |S \ C|
        foreach u ∈ S \ C do
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        C ← C ∪ S
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    return S'                                  // Cover of U
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Reminder: $\sum_{u \in U} \text{price}(u)$ completely pays for $\mathcal{S}'$.

New: LP-based Analysis

Observation. For each $u \in U$, $\text{price}(u)$ is a dual variable

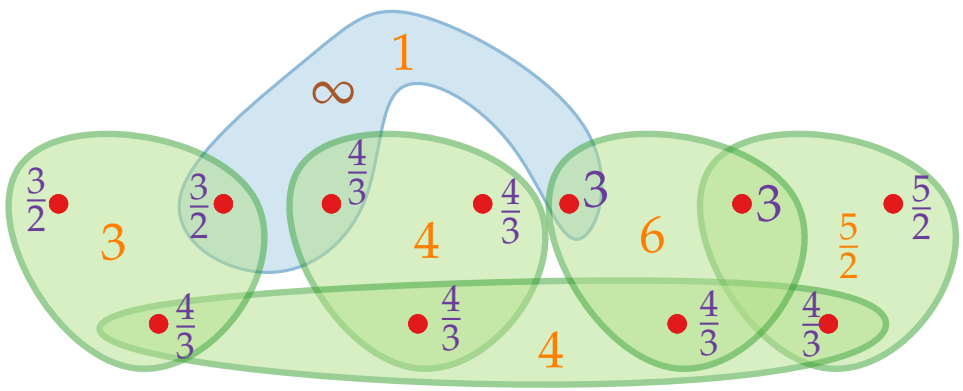
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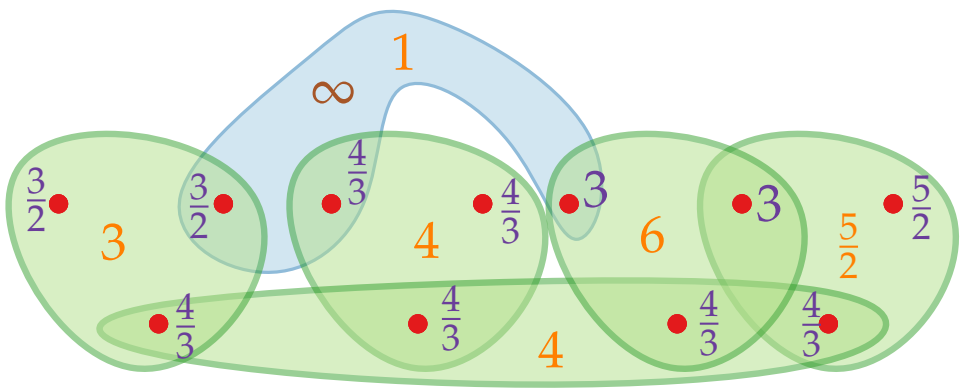


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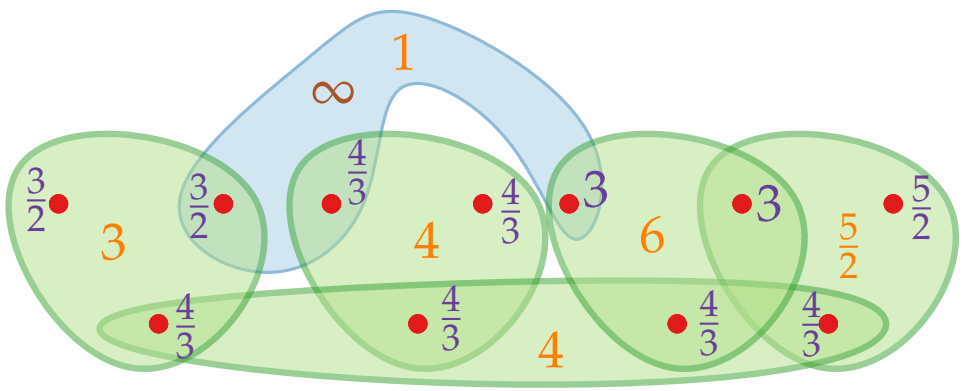


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Scale duale variables such that no set is overpacked.



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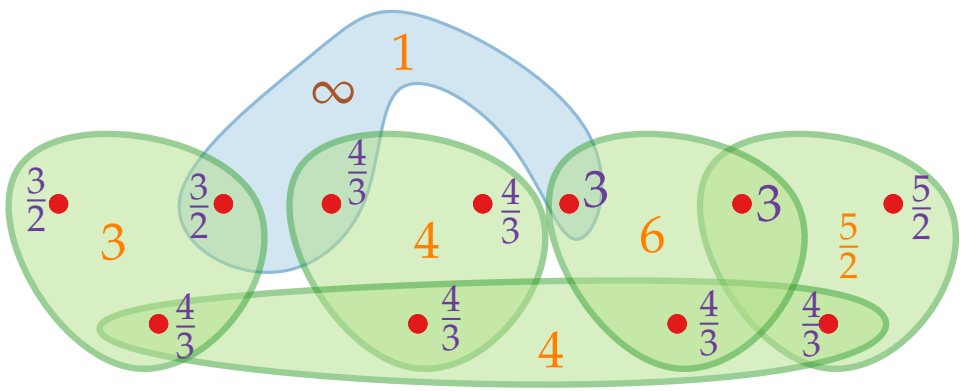
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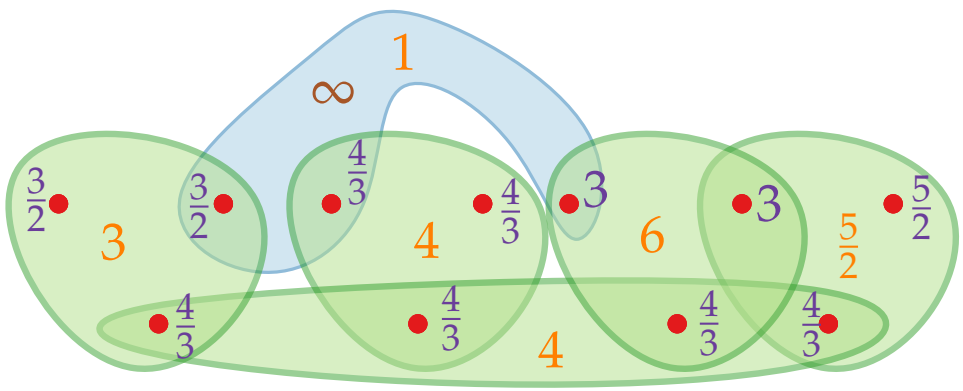
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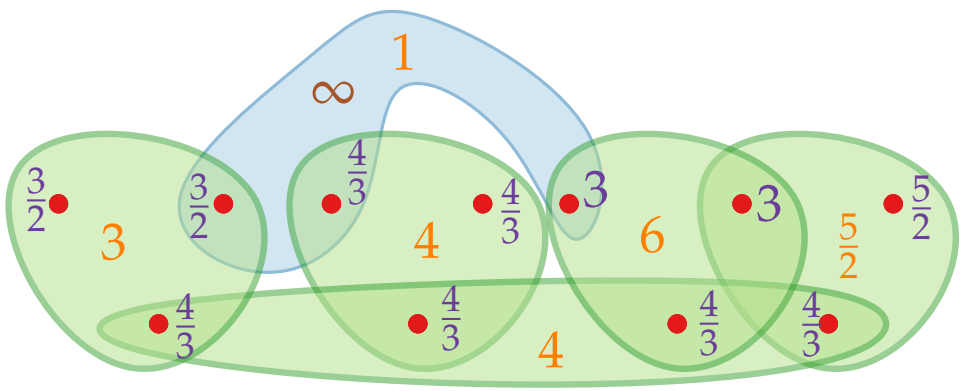
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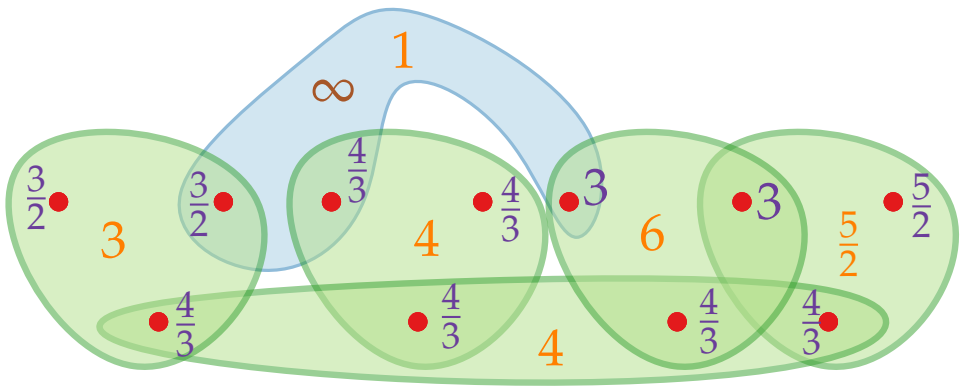
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Theorem. GreedySetCover is a factor-$\mathcal{H}_k$-approximation algorithm for SetCover, where $k = \max_{S \in \mathcal{S}} |S|$.

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Dual solution allows a *per-instance* estimation

. . . which might be stronger than worst-case bound $\mathcal{H}_k$